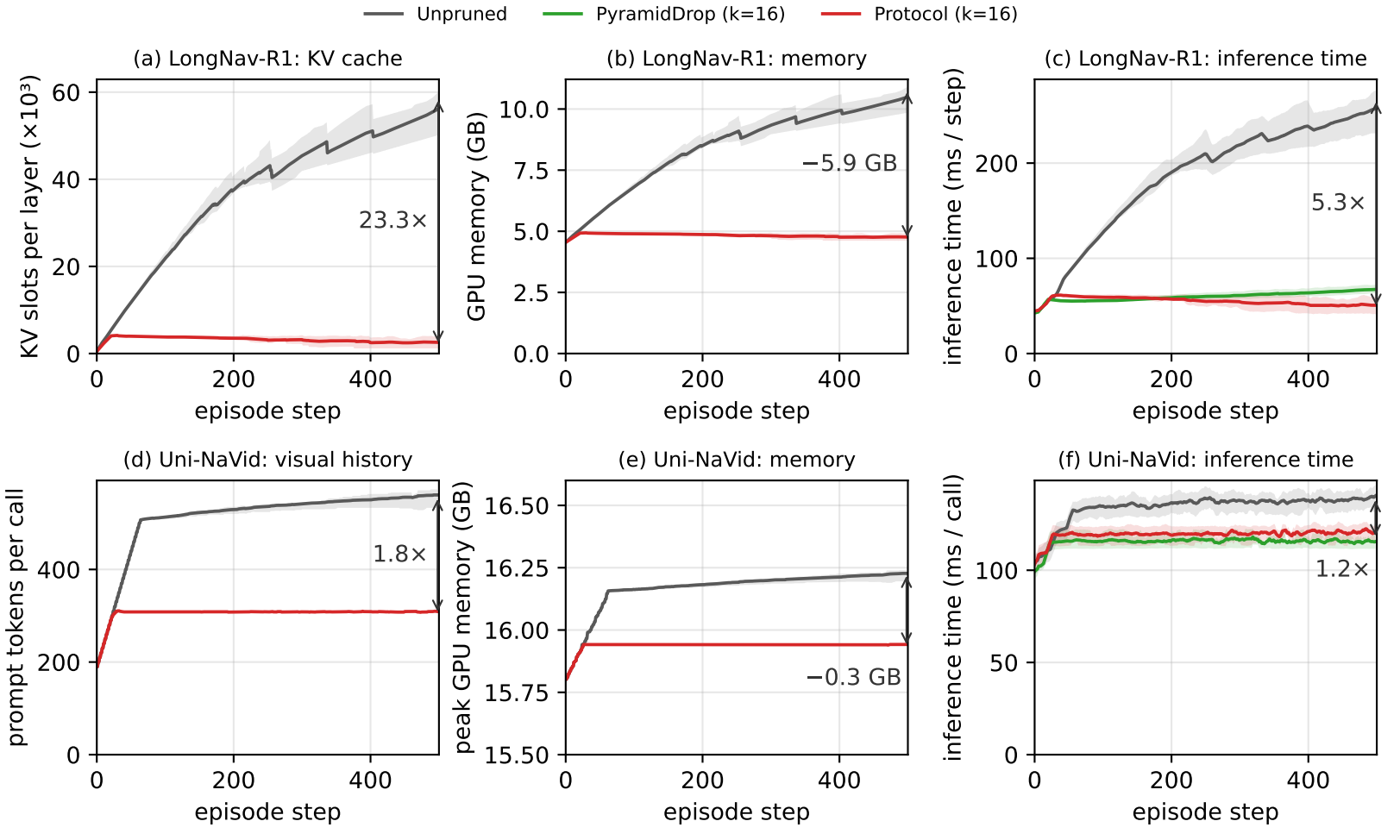

Unpruned
PyramidDrop (k=16)
Protocol (k=16)
(a) LongNav-R1: KV cache
KV slots per layer (×10³)
23.3×
episode step
(b) LongNav-R1: memory
GPU memory (GB)
−5.9 GB
episode step
(c) LongNav-R1: inference time
inference time (ms / step)
5.3×
episode step
(d) Uni-NaVid: visual history
prompt tokens per call
1.8×
episode step
(e) Uni-NaVid: memory
peak GPU memory (GB)
−0.3 GB
episode step
(f) Uni-NaVid: inference time
inference time (ms / call)
1.2×
episode step